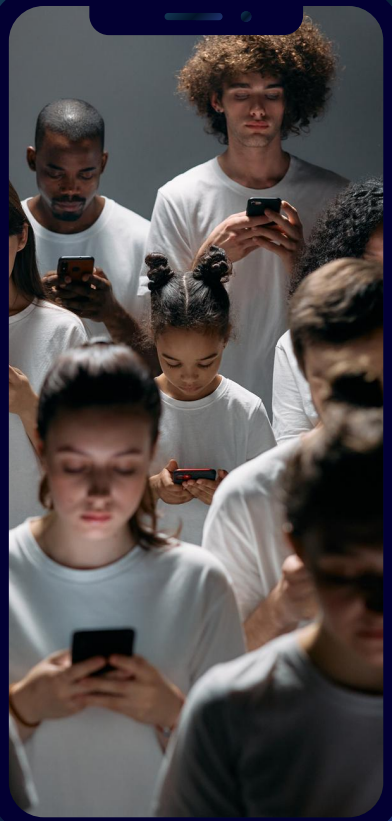


Predicting Customer Churn at SyriaTel

A Machine Learning Strategy for Proactive Customer Retention

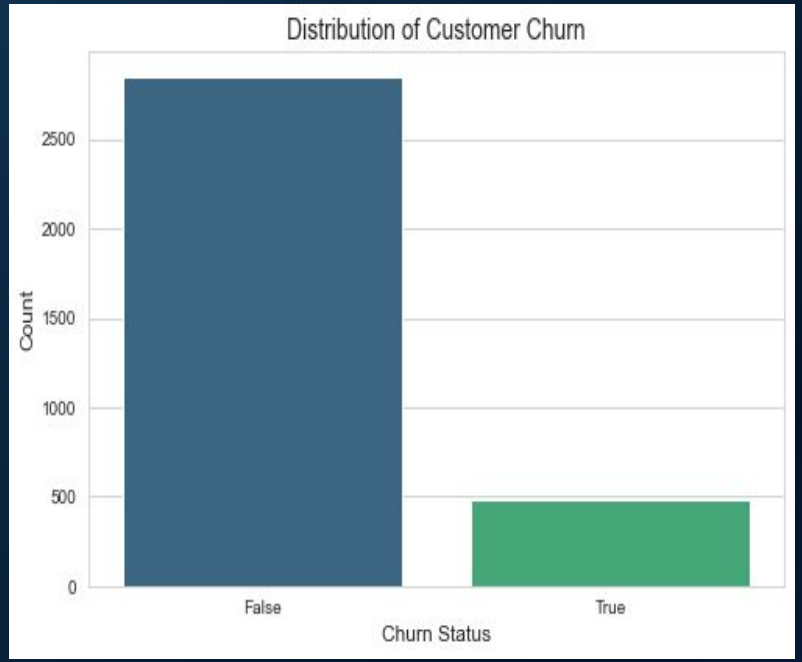
Customer churn represents a major revenue risk for telecommunications providers.

This project develops a predictive early-warning system that identifies customers likely to leave before they churn.

The goal is to enable timely, targeted retention interventions that improve customer lifetime value.

| Presented by: Daniella Muli
| Moringa School Data Science
| March 2026

Why Churn Prediction Matters



Approximately 14.5% of customers in the dataset have churned, representing a significant loss of recurring revenue.

Retaining an existing customer is substantially more cost-effective than acquiring a new one.

Without predictive insights, retention efforts are often reactive and inefficient.

Project Goal:
Build a model capable of identifying at least 65% of churners in advance, allowing SyriaTel to intervene strategically.

Customer Behavioral Data as a Signal

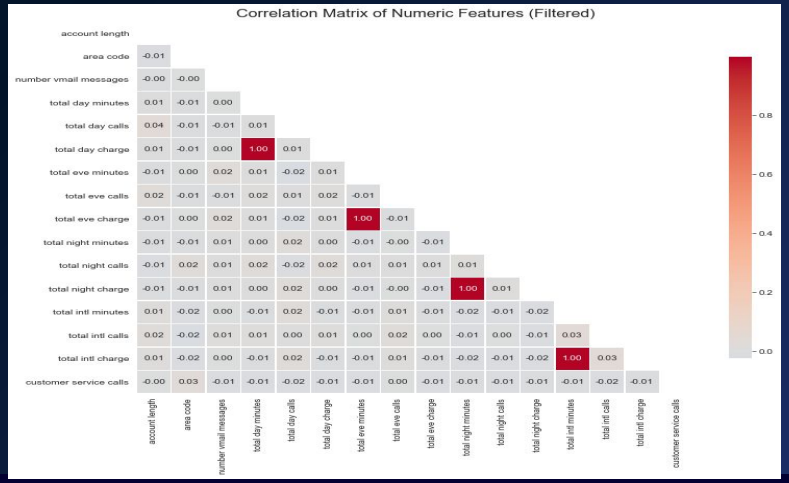
Dataset contains 3,333 customers with usage patterns, service interactions, and plan subscriptions.

Key preprocessing steps included:

Removing unique identifiers (e.g., phone numbers)

Handling perfect multicollinearity between usage minutes and corresponding charges

Creating engineered features such as total usage intensity



Insight:
Customer behavior — not demographics — provides the strongest signal of churn risk.

Behavioral Drivers of Churn

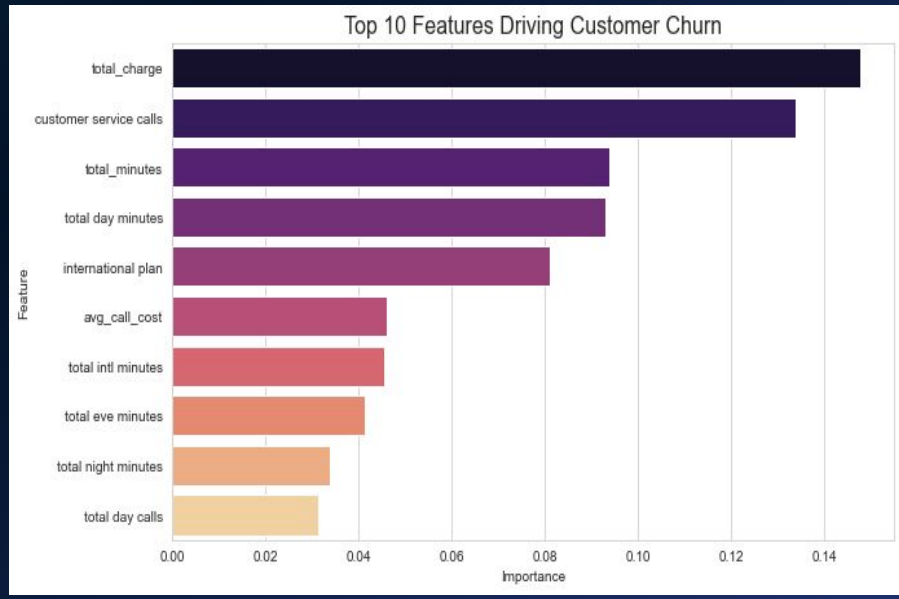
Model insights reveal several high-impact predictors:

High daytime usage: Customers with heavy peak-hour usage are more likely to churn, likely due to pricing sensitivity.

Frequent customer service calls: Churn probability increases significantly after multiple support interactions.

International plan subscription: Indicates possible dissatisfaction with pricing or perceived value.

Interpretation:
Churn is often driven by cost pressure and service friction.



Building a Prediction System

How the Ensemble Works:

Logistic Regression

Detects consistent linear patterns in customer behaviour

Decision Tree

Identifies critical tipping points such as repeated service calls

Random Forest

Captures complex usage patterns across multiple variables

⭐ Voting Ensemble (Final Decision)

Final churn probability is determined by combining predictions from all models.

To accurately identify customers at risk of churn, we evaluated multiple machine learning approaches with different strengths.

Each model captures a unique aspect of customer behaviour:

1. Logistic Regression identifies consistent trends across large customer segments.
2. Decision Trees detect critical behavioural tipping points such as repeated support calls.
3. Random Forest uncovers complex patterns across usage intensity, plans, and service interactions.

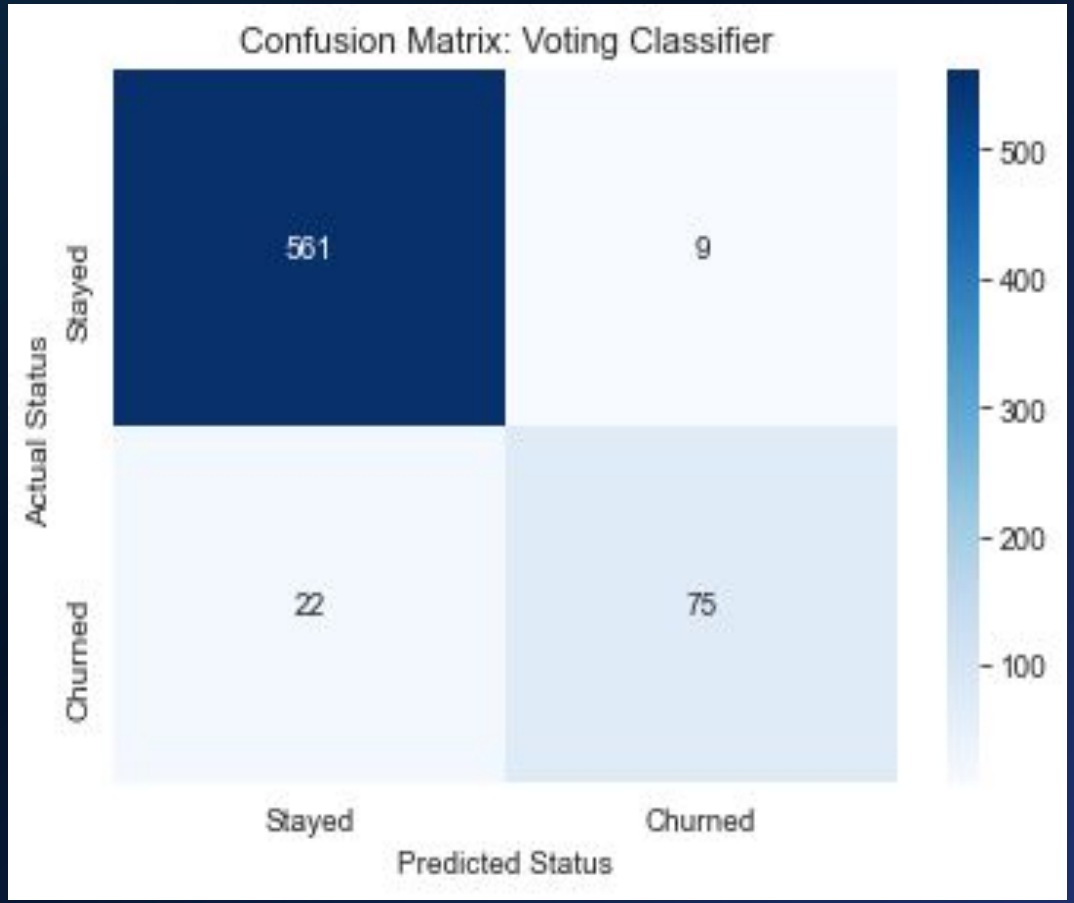
To maximize predictive reliability, these models were combined into a Soft Voting Ensemble, allowing SyriaTel to make more confident retention decisions.

Model Performance & Reliability

- Final Voting Classifier performance on unseen test data:

- Accuracy: 95.3%
- Recall (Churn): 77.3%
- Precision (Churn): 89.3%
- F1-Score: 82.9%

- This means the model correctly identifies approximately 3 out of every 4 customers who are likely to churn.



Business Interpretation:
The model provides a reliable targeting mechanism for retention campaigns.

Recommendations

1. Proactive Retention Outreach

Score customers regularly and engage high-risk individuals early.

2. Customer Service Experience Improvement

Escalate customers with repeated support interactions to specialized retention teams.

3. Usage-Based Plan Optimization

Introduce tailored bundles for high daytime users to reduce price sensitivity.

4. International Plan Redesign

Enhance perceived value through pricing adjustments or bundled benefits.

Model Limitations & Future Opportunities

Class Imbalance

- Only 14.5% of customers churn, making it inherently difficult to identify every at-risk subscriber.
- Future improvements may include collecting more churn data and applying advanced resampling or threshold optimization.

Limited Feature Availability

- Important predictors such as network quality metrics, customer sentiment, and competitor pricing were not available.
- Integrating these signals could significantly improve predictive accuracy.

Static Snapshot of Behaviour

- The dataset represents customer activity at a single point in time.
- However, churn drivers can evolve due to pricing changes, competition, or seasonal usage trends.
- Regular model retraining and performance monitoring will be essential.

High Geographic Dimensionality

- State-level one-hot encoding increases feature complexity and may reduce model efficiency.
- Future iterations could explore regional clustering or more compact encoding strategies.

Implementation Roadmap

- Pilot Phase: Deploy model scoring on a small customer segment.
- Performance Monitoring: Track retention rate improvements and campaign ROI.
- Model Refresh: Retrain periodically as usage patterns evolve.
- Data Expansion: Integrate network quality, sentiment, and competitive pricing data.

Conclusion

This project demonstrates how machine learning can transform churn management from reactive to proactive.

By identifying at-risk customers early, SyriaTel can:

Reduce revenue loss

Improve customer satisfaction

Strengthen long-term profitability

Thank You!

Questions?

Presented by Daniella Muli